



DIPARTIMENTO DI ELETTRONICA,
INFORMAZIONE E BIOINGEGNERIA

Politecnico di Milano

Machine Learning (Code: 097683)

September 10, 2022

Surname:

Name:

Student ID:

Row:

Column:

Time: 2 hours

Maximum Marks: 33

- The following exam is composed of **8 exercises** (one per page). The first page needs to be filled with your **name, surname and student ID**. The following pages should be used **only in the large squares** present on each page. Any solution provided either outside these spaces or **without a motivation** will not be considered for the final mark.
- During this exam you are **not allowed to use electronic devices**, such as laptops, smartphones, tablets and/or similar. As well, you are not allowed to bring with you any kind of note, book, written scheme, and/or similar. You are also not allowed to communicate with other students during the exam.
- The first reported violation of the above-mentioned rules will be annotated on the exam and will be considered for the final mark decision. The second reported violation of the above-mentioned rules will imply the immediate expulsion of the student from the exam room and the **annulment of the exam**.
- You are allowed to write the exam either with a pen (black or blue) or a pencil. It is your responsibility to provide a readable solution. We will not be held accountable for accidental partial or total cancellation of the exam.
- The exam can be written either in **English** or **Italian**.
- You are allowed to withdraw from the exam at any time without any penalty. You are allowed to leave the room not early than half the time of the duration of the exam. You are not allowed to keep the text of the exam with you while leaving the room.
- **Three of the points will be given on the basis on how quick you are in solving the exam.** If you finish earlier than 30 min before the end of the exam you will get 3 points, if you finish earlier than 20 min you will get 2 points and if you finish earlier than 10 min you will get 1 point (the points cannot be accumulated).
- The box on Page 10 can only be used to complete the Exercises 7, and/or 8.

Ex. 1	Ex. 2	Ex. 3	Ex. 4	Ex. 5	Ex. 6	Ex. 7	Ex. 8	Time	Tot.
/ 7	/ 7	/ 2	/ 2	/ 2	/ 2	/ 4	/ 4	/ 3	/ 33

Exercise 1 (7 marks)

Describe **Support Vector Machines** (SVMs) for supervised classification problems. In particular explain how do they work, their strengths and weaknesses.

Exercise 2 (7 marks)

Describe the **PCA** technique and what it is used for.

Exercise 3 (2 marks)

Consider the following snippet of code:

```
1 V1 = np.linalg.inv(np.eye(nS) - gamma * pi @ P_sas) @ (pi @ R_sa)
2
3 V_old = np.zeros(nS)
4 tol = 0.0001
5 V2 = pi @ R_sa
6 while np.any(np.abs(V_old - V2) > tol):
7     V_old = V2
8     V2 = pi @ (R_sa - gamma * P_sas @ V)
```

1. Describe the purpose of the procedure of line 1 and the purpose of the procedure of lines 3-8. Are they correct? If not, propose a correction.
2. What is the main disadvantage of the procedure of line 1 compared to the one of lines 3-8?
3. What happens to the two procedures when `gamma=1`?

1. The procedure of line 1 computes the closed-form solution of the state value function V^π of policy `pi` in the MDP with transition model `P_sas`, reward `R_sa` and discount factor `gamma`. The procedure of lines 3-8 performs the iterative application of the Bellman expectation operator to compute the same value function V^π . The iterative procedure is stopped when a given threshold `tol` between consecutive approximation is reached. However, line 8 does contain a mistake and should be corrected as follows:
`V2 = pi @ (R_sa + gamma * P_sas @ V).`
2. The main disadvantage of the procedure of line 1 compared to the one of lines 3-8 is a computational one, i.e., the computation of the closed-form solution might be infeasible when the number of states/actions is large.
3. When `gamma=1` the procedure of line 1 might lead to a singular matrix (attempting to invert it), whereas the procedure of lines 3-8 might never reach the requested tolerance `tol`.

Exercise 4 (2 marks)

Tell if the following statements about bias-variance trade-off are true or false. Motivate your answers. Consider a regression problem with input variables x_1 , x_2 , and x_3 , that are linearly independent.

1. In linear regression, if we replace variable x_1 with $x_1 + x_2$, we do not change the bias of the model.
2. In linear regression, if we replace variable x_1 with $x_1^2/100$, we might increase the variance of the model.
3. For an arbitrary model, if we remove variable x_2 , we do not increase the variance of the model.
4. For an arbitrary model, if we add variable x_3^2 , we cannot increase the bias of the model.

1. TRUE. Indeed, since the variables are linearly independent, the linear models representable with $\{x_1, x_2, x_3\}$ are the same as the one representable with $\{x_1 + x_2, x_2, x_3\}$.
2. TRUE. It might be the case that the variance is larger, since the linear models representable with $\{x_1, x_2, x_3\}$ are not comparable with the ones representable with $\{x_1^2/100, x_2, x_3\}$.
3. TRUE. Indeed, regardless the used model, the space of representable models with a subset of the variables, i.e., $\{x_1, x_3\}$ is contained in the space of representable models with all the variables.
4. TRUE. Indeed, adding variables enlarges (or, possibly, leaves unchanged) the space of representable models.

Exercise 5 (2 marks)

Tell if the following functions are valid kernels. Motivate your answers. Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$:

1. $k_1(\mathbf{x}, \mathbf{y}) = \mathbf{x}^T \mathbf{y} + \mathbf{x}^T \mathbf{1} + \mathbf{y}^T \mathbf{1} + d$, where $\mathbf{1} \in \mathbb{R}^d$ is the vector of all ones.
2. $k_2(\mathbf{x}, \mathbf{y}) = \mathbf{x}^T \mathbf{y} - \|\mathbf{x}\|^2$
3. $k_3(\mathbf{x}, \mathbf{y}) = k_1(\cos(\mathbf{x}), \cos(\mathbf{y}))^3$, where the $\cos(\cdot)$ function is applied element-wise.
4. $k_4(\mathbf{x}, \mathbf{y}) = \exp(k_2(\mathbf{x}, \mathbf{y}) + k_2(\mathbf{y}, \mathbf{x}))$

1. TRUE. By definition, selecting $\phi(\mathbf{x}) = \mathbf{x} + \mathbf{1}$, we have $k_1(\mathbf{x}, \mathbf{y}) = \phi(\mathbf{x})^T \phi(\mathbf{y})$.
2. FALSE. $k_2(\mathbf{x}, \mathbf{y})$ is not symmetric.
3. TRUE. Since we are considering the same transformation $\cos(\cdot)$ applied to both arguments, then $(\cdot)^3$ is a polynomial transformation with non-negative coefficients, and k_1 is a kernel.
4. TRUE. With simple computations, we obtain:

$$k_4(\mathbf{x}, \mathbf{y}) = \exp(-k_2(\mathbf{x}, \mathbf{y}) + k_2(\mathbf{y}, \mathbf{x})) = \exp(2\mathbf{x}^T \mathbf{y} - \|\mathbf{x}\|^2 - \|\mathbf{y}\|^2) = \exp(-\|\mathbf{x} - \mathbf{y}\|^2)$$

that is the Gaussian kernel with $\sigma^2 = \frac{1}{2}$.

Exercise 6 (2 marks)

A pool has the problem that many customers are not wearing bathing caps, which is against the rules. The currently implemented method to prevent such a behaviour is that the lifeguards are checking the pool and notify to the person not wearing the bathing cap that she/he is in violation of the pool rules. However, this task is boring and prone to errors (lifeguards are commonly distracted by other issues). You are required to design a new system to automatically detect such a violation.

1. Provide a solution to tell if at least one person is not wearing a bathing cap. Define the data you are required to collect, the type of problem you are solving and suggest a method that is suitable for the situation.
2. Provide a solution if, instead, you want to count the number of people not wearing a bathing cap. Does this problem requires new data? Is it a different problem? Are you required to use a different technique?

1. The solution would be to install a camera on the ceiling of the pool and use the images as input for a classifier. This process would require the acquisition of a dataset of images with the corresponding labels (there is someone in violation of the pool rules/ everything is fine). A common way to solve classification problems involving images is to use CNN.
2. In this case we would require a different label for the images. We need to couple each image with the number of people not wearing a bathing cap. In this case we are solving a regression problem. Assuming we are able to extract a set of meaningful features from the images, we might resort to any regression method (such as Linear Regression). Otherwise, we can still use CNN to solve the regression problem directly from the images.

Exercise 7 (4 marks)

Consider an initial parameter for a Linear Regression $\mathbf{w}^{(0)} = [0 \ 1 \ 0]^\top$ and a loss function of the form:

$$J(\mathbf{w}) = \frac{1}{2N} \sum_{n=1}^N (\mathbf{w}^\top \mathbf{x}_n - t_n)^2,$$

where N is the number of data.

1. Given a learning rate of $\alpha = 0.2$, derive the update given from the gradient descent algorithm to minimize the loss when it is applied to single data;
2. Apply sequentially the above rule on the following dataset (one at a time):

$$\mathbf{x}_1 = [1 \ 3 \ 2]^\top, \ t_1 = 4, \quad \mathbf{x}_2 = [1 \ 1 \ 1]^\top, \ t_2 = 1, \quad \mathbf{x}_3 = [5 \ 10 \ 5]^\top, \ t_3 = 14,$$

3. Do we have any issue about using the gradient descend algorithm if we use the following loss function instead?

$$J(\mathbf{w}) = \frac{1}{2N} \sum_{n=1}^N |\mathbf{w}^\top \mathbf{x}_n - t_n|$$

1. The loss for a single data is $J(\mathbf{w}) = \frac{1}{2}(\mathbf{w}^\top \mathbf{x}_n - t_n)^2$ and the gradient of the loss becomes $\nabla J(\mathbf{w}) = (\mathbf{w}^\top \mathbf{x}_n - t_n)\mathbf{x}_n$. The update of the gradient descent methods is then:

$$\mathbf{w}_{k+1} \leftarrow \mathbf{w}_k - \alpha \nabla J(\mathbf{w}_k) = \mathbf{w}_k - 0.2(\mathbf{w}_k^\top \mathbf{x}_k - t_k)\mathbf{x}_k$$

- 2.

$$\mathbf{w}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} - 0.2 \cdot (-1) \cdot \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 0.2 \\ 1.6 \\ 0.4 \end{bmatrix}$$

$$\mathbf{w}_2 = \begin{bmatrix} 0.2 \\ 1.6 \\ 0.4 \end{bmatrix} - 0.2 \cdot (1.2) \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -0.04 \\ 1.36 \\ 0.16 \end{bmatrix}$$

$$\mathbf{w}_3 = \begin{bmatrix} -0.04 \\ 1.36 \\ 0.16 \end{bmatrix} - 0.2 \cdot (0.2) \cdot \begin{bmatrix} 5 \\ 10 \\ 5 \end{bmatrix} = \begin{bmatrix} -0.24 \\ 0.96 \\ -0.04 \end{bmatrix}$$

3. Since the provided loss function has a gradient almost everywhere, we can apply such a strategy to find the optimal solution even in this case. The only issue is if we have parameter vectors s.t. $|\mathbf{w}^\top \mathbf{x}_n - t_n| = 0$.

Exercise 8 (4 marks)

Suppose we collect data for a group of customers of a car insurance company with variables age x_1 , number of accidents occurred so far x_2 and the probability to have a car accident in the next year t . We fit a logistic regression and produce estimated coefficients: $w_0 = -6$, $w_1 = 0.05$ and $w_2 = 0.5$.

1. Compute the probability that a 20-years old customer who did 4 accidents will have a new accident in the next year (provide the formula and substitute the values, leave the calculations).
2. How many accidents would a 30-years old customer need to have to have 50% chance of getting another one in the next year?
3. Do you think that the parameters learned by the model are meaningful w.r.t. the problem it addresses?
4. Do you think that this model is missing some crucial information about the customer? Do you need to retrain the model from scratch in the case you added a new variable x_3 to it?

1. The probability is equal to:

$$P(y = 1|\mathbf{x}) = \frac{1}{1 + e^{-(w_0 + w_1 x_1 + w_2 x_2)}} = \frac{1}{1 + e^{-(-6 + 20 \cdot 0.05 + 4 \cdot 0.5)}} = \frac{1}{1 + e^3}$$

2. On the boundary of the Logistic Regression we have a 50% chance to get an accident in the next year, therefore:

$$w_0 + w_1 x_1 + w_2 x_2 = 0 \rightarrow -6 + 1.5 + 0.5 x_2 = 0 \rightarrow x_2 = 9$$

3. It seems to be correct that the more the accidents I had in the past the more the chance I will have a new accident. Conversely an increase of the age should decrease the chance of having an accident since drivers get skilled over time, reducing the chance of having a new accident.
4. It seems that the features included are meaningful w.r.t. the problem solved. Probably we are missing the information about the amount of time the customer had the insurance, since old customers have more accidents only because they have a longer period over which they had the possibility to damage their car. Since there is no easy way of decoupling the loss function w.r.t. a single feature, one has to retrain everything from scratch.

